

Analysis of Competing Solutions: Ethereum-Based and Centralized Systems for Content Rights Management

This analysis evaluates competing solutions for cross-chain content rights management, with a focus on decentralized subscription, pay-per-view (PPV), and purchase models. It references recent literature and industry implementations to assess Ethereum-based systems—often utilizing smart contracts for blockchain-native digital rights management (DRM) and centralized systems such as Adobe Content Server and Netflix’s backend. The evaluation criteria align with the thesis’s key metrics: scalability (e.g., transaction throughput and load handling), interoperability (e.g., cross-platform and network compatibility), and economic efficiency (e.g., fees, revenue sharing, and cost savings). Ethereum-based solutions highlight decentralization but encounter network-specific limitations, whereas centralized systems prioritize speed and control at the expense of transparency.

The analysis identifies deficiencies, such as Ethereum’s single-chain bottlenecks and the intermediary dependencies inherent to centralized systems, which the proposed Polkadot-based framework alleviates through multi-chain interoperability and shared security for resources.

Ethereum-Based Solutions

Ethereum-based DRM systems integrate smart contracts and non-fungible tokens (NFTs) to enable programmable rights enforcement and monetization. These systems leverage Ethereum’s ecosystem for tokenization and automation; however, they are often limited to Ethereum Virtual Machine (EVM)-compatible blockchains, thereby reducing inherent cross-chain interoperability functionalities.

Key Example 1: Elacity Web3 DRM

Elacity is a blockchain-integrated Digital Rights Management (DRM) platform that utilizes Non-Fungible Tokens (NFTs) for the management of video and multimedia assets. It accommodates various business models, including subscription services (providing ongoing access through tokenized memberships), Pay-Per-View (PPV), which involves one-time payments for specific views with access tokens that are time-limited, and outright purchases, which confer full ownership transferred as NFTs. The platform employs smart contracts to automate revenue distribution to content creators, typically retaining between 95–98%, and to enforce content scarcity and encryption measures. All transactions are recorded on a decentralized ledger to ensure transparency and security transparency.

- **Scalability:** Handles expansion to diverse assets (e.g., audio, 3D models) via Web3 protocols; however, Ethereum’s gas fees and network congestion may restrict high-volume pay-per-view (PPV) during operations peaks.
- **Interoperability:** Collaborates with standards such as MPEG-21 and the Metaverse Forum to ensure compatibility with traditional tools, thereby facilitating hybrid utilization; however, it is primarily centered on Ethereum, necessitating bridges for non-EVM systems chains.
- **Economic Efficiency:** Reduces intermediary fees, such as those exemplified by YouTube’s 45% share, with immediate payouts. This approach facilitates collaborative revenue

models, including cross-creator royalties. Additionally, gas costs, approximately \$0.01 to \$0.10 per transaction, are relatively insignificant overhead.

Key Example 2: UPV Blockchain DRM Framework

This permissioned blockchain framework, inspired by Ethereum’s Swarm for storage and Ether for payments, aims to facilitate music streaming through an “Asset Assertion (AA)” token system. It integrates cross-platform players within content files to monitor usage and enforce access controls via smart contracts. Monetization mechanisms encompass subscriptions (recurring token payments), pay-per-view (PPV) (per-stream fees), and purchases (ownership tokens). All interactions, such as streams and royalties, are reconciled to ensure accuracy and transparency on-chain.

- **Scalability:** Permissioned nodes facilitate rapid consensus suitable for high-volume streaming; whitelisting distributors endorses enterprise-scale operations without reliance on public blockchains bottlenecks.
- **Interoperability:** AA deploys self-install readers across multiple devices, integrating with various cryptocurrencies including AA, Ether, and Bitcoin. It supports multi-party models but necessitates a custom token adoption.
- **Economic Efficiency:** Automates royalty payments (e.g., immediate payments compared to months in traditional systems), thereby decreasing piracy-related losses (~\$12 billion annually in the music industry). Listener rewards generate new revenue streams, although token volatility presents challenges risks.

Overall Strengths and Limitations

Ethereum-based systems excel in transparency and creator control, aligning with Web3 principles. However, scalability is constrained by Ethereum’s approximately 15 transactions per second (TPS), compared to Polkadot’s parachain potential of over 1,000 TPS. Furthermore, interoperability often necessitates external bridges, which can increase latency to between 2 and 10 seconds. Economically, these systems reduce fees to less than 5%, but they incur variable gas costs.

Centralized Systems

Centralized Digital Rights Management (DRM) depends on proprietary servers, encryption standards (such as Widevine, PlayReady, and FairPlay), and license servers to regulate access. These technologies are predominant in over-the-top (OTT) services (e.g., Netflix) and enterprise publishing (e.g., Adobe), emphasizing reliability over other considerations decentralization.

Key Example 1: Adobe Content Server (Primetime DRM)

Adobe’s solution safeguards PDFs, videos, and eBooks through server-side encryption and license management. For subscriptions, such as Adobe Creative Cloud, it ensures ongoing access; PPV utilizes time-limited licenses; and purchases provide perpetual keys. Integrated with AWS for worldwide distribution, it facilitates support across multiple devices policies.

- **Scalability:** Handles millions of users (e.g., enterprise deployments); cloud scaling via AWS achieves approximately 5,000 TPS.
- **Interoperability:** Compatible with iOS and Android platforms through FairPlay and Widevine; integrates with Content Management Systems such as WordPress. However, vendor lock-in constraints limit customization.
- **Economic Efficiency:** The fee is approximately \$0.05 per API call; this measure provides protection against piracy, thereby safeguarding millions in potential losses. However, platform commissions ranging from 20% to 30% diminish earnings for creators shares.

Key Example 2: Netflix Backend DRM

Netflix employs multi-DRM technologies (Widevine for Android/Chrome, PlayReady for Xbox, FairPlay for Apple) accompanied by centralized license servers to manage encrypted streams. Subscriptions grant unlimited access and support offline caching; pay-per-view (e.g., events) imposes session limits; purchases are infrequent but are facilitated through proprietary content libraries. Analytics monitor usage to enable dynamic adjustments pricing.

- **Scalability:** Processes billions of streams daily on AWS; adaptive bitrate manages peak loads with a 99.99% reliability rate uptime.
- **Interoperability:** Standards such as CMAF and DASH facilitate cross-device playback; EME APIs support browser integration, although ecosystem silos created by Apple and Google hinder seamless operation enforcement.
- **Economic Efficiency:** The platform offers low per-stream costs estimated at approximately \$0.01 to \$0.02 through scaling. It features transparent revenue sharing, with creators receiving between 50% and 70%. However, it incurs substantial infrastructure expenses, estimated at around \$1 billion annually, and involves a 30% intermediary fee.

Overall Strengths and Limitations

Centralized systems provide low latency (less than 200 milliseconds) and high reliability, making them suitable for mass-market subscriptions. Scalability is strongly supported through cloud infrastructure; however, interoperability issues are prevalent due to ecosystem silos, such as FairPlay's exclusivity with Apple. Economically, these systems reduce initial costs but entail significant ongoing fees and lack transparency in revenue distribution splits.

Comparative Analysis

The table below presents a comparison among Ethereum-based, centralized, and the proposed Polkadot system across various thesis metrics. The data is derived from benchmarks, such as Ethereum's TPS from academic literature and centralized systems from providers like AWS and Netflix reports).

Table 1: Comparison of Ethereum-Based, Centralized, and Proposed Polkadot Framework

Metric	Ethereum-Based (e.g., Elacity/UPV)	Centralized (e.g., Adobe/Netflix)	Proposed Polkadot Framework
Scalability (TPS)	Low (15–100 TPS); congestion-prone	High (1,000–5,000 TPS); cloud-optimized	High (500–1,000+ TPS via parachains); shared security
Interoperability	Medium (EVM bridges needed; 2–10s latency)	High within ecosystems (e.g., CMAF); silos across	Excellent (native XCM for multi-chain; <2s transfers)
Economic Efficiency	High creator retention (95%+); gas fees (\$0.01–0.10)	Medium (50–70% shares); infra costs (\$0.01–0.05/call)	Superior (<5% fees); on-chain transparency, Monte Carlo savings (60% vs. centralized)
Subscription Support	Tokenized recurring; automated but volatile	Seamless caching; reliable but opaque	XCM-activated recurring; low-latency verification
PPV/Purchase	NFT/time-bound tokens; instant but chain-bound	Session licenses; fast but revocable centrally	Cross-chain tokens; ownership via bridges (e.g., Ethereum)
Key Trade-off	Decentralization vs. speed	Control vs. transparency	Balances via shared model; addresses gaps in both

Blockchain technology, exemplified by Ethereum, surpasses centralized systems in transparency and security, owing to immutable ledgers in contrast to susceptible servers. However, it falls short in scalability and efficiency due to the constraints inherent in single-chain architectures. Centralized systems excel in user experience but compromise creator control through fees and opacity. The heterogeneous chains of Polkadot address Ethereum’s siloed limitations, providing enhanced cross-network pay-per-view and subscription services both.

Conclusion and Implications for Thesis

Competing solutions reveal inherent trade-offs: Ethereum-based platforms facilitate innovative monetization strategies but face challenges with cross-chain scalability. Conversely, centralized solutions offer greater efficiency at the expense of decentralization. The proposed framework employs ink! contracts and XCM to bridge these gaps, aiming for a 99% success rate and a Herfindahl-Hirschman Index (HHI) below 1,500 to ensure balanced decentralization. Future evaluation, scheduled for Phase 4, should benchmark these solutions against Elacity (Ethereum) and Adobe (centralized) utilizing simulated load tests. This analysis further enhances the related work section, emphasizing the novelty of Polkadot’s multi-chain framework for achieving economic efficiency in content management rights.

Implementation Notes (Phase 4–5 Measured Results)

The comparative table above was compiled during Phase 1 using theoretical estimates and published literature. Phase 4–5 implementation and testing produced measured results that should replace the theoretical projections in the thesis:

Table 2: Phase 1 Estimates vs Phase 5 Measured Results

Metric	Phase 1 Estimate	Phase 5 Measured	Notes
Polkadot TPS	500–1,000+	27 TPS sustained, 283 TPS theoretical	Single parachain throughput is limited by relay chain slot allocation (~6s blocks). 27 TPS = 2.3M ops/day. Horizontal scaling occurs via additional parachains, not within a single chain.
XCM transfer latency	<2 seconds	18–32 seconds (3–5 parachain blocks)	Theoretical estimate did not account for HRMP relay hops through the relay chain
Economic efficiency	<5% fees, Monte Carlo 60% savings	<1% blockchain fees; centralized comparison shows 270× performance cost for trustlessness	Monte Carlo simulation not implemented; comparative analysis used a real centralized benchmark instead
Interoperability	“Excellent, native XCM”	XCM proven (100% success); Snowbridge E2E proven (2 ETH bridged)	Confirmed — cross-chain and cross-ecosystem both demonstrated
Success rate target	99% success, HHI < 1,500	100% XCM success; 100% local extrinsic success (at tested concurrency)	Exceeded the success rate target

Updated Comparative Table (with measured data)

Table 3: Updated Comparison with Measured Polkadot Results

Metric	Ethereum-Based	Centralized	Polkadot (Measured)
Scalability (TPS)	15–100 TPS	1,000–5,000 TPS	27 TPS sustained; 283 TPS theoretical per parachain
Interoperability	EVM bridges (2–10s)	Ecosystem silos	XCM: 18–32s; Snowbridge: ~1 min after finalization
Economic Efficiency	95%+ creator retention	50–70% creator share	100% to creator (self-publishing); configurable royalty splits
Subscription	Tokenized; volatile	Seamless; opaque	On-chain with auto-renewal via on_initialize hook
PPV/Purchase	NFT tokens; chain-bound	Session licenses; revocable	Counter-based PPV; permanent ownership with cross-chain transfer
Trust Model	Trustless (single chain)	Trusted third party	Trustless (relay chain consensus + Snowbridge light client)
Latency	12–15s (Ethereum L1)	<200ms	~6s (1 block), 12–18s finality